

MdGH3.6 is targeted by MdMYB94 and plays a negative role in apple water-deficit stress tolerance

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SUMMARY

Drought significantly limits apple fruit production and quality. Decoding the key genes involved in drought stress tolerance is important for breeding varieties with improved drought resistance. Here, we identified GRETCHEN HAGEN3.6 (GH3.6), an indole-3-acetic acid (IAA) conjugating enzyme, to be a negative regulator of water-deficit stress tolerance in apple. Overexpressing *MdGH3.6* reduced IAA content, adventitious root number, root length and water-deficit stress tolerance, whereas knocking down *MdGH3.6* and its close paralogs increased IAA content, adventitious root number, root length and water-deficit stress tolerance. Moreover, *MdGH3.6* negatively regulated the expression of wax biosynthetic genes under water-deficit stress and thus negatively regulated cuticular wax content. Additionally, *MdGH3.6* negatively regulated reactive oxygen species scavengers, including antioxidant enzymes and metabolites involved in the phenylpropanoid and flavonoid pathway in response to water-deficit stress. Further study revealed that the homolog of transcription factor AtMYB94, rather than AtMYB96, could bind to the *MdGH3.6* promoter and negatively regulated its expression under water-deficit stress conditions in apple. Overall, our results identify a candidate gene for the improvement of drought resistance in fruit trees.

Keywords: apple (*Malus × domestica*), water-deficit stress, GRETCHEN HAGEN3 (GH3) genes, *MdGH3.6*, *MdMYB94*.

INTRODUCTION

Under natural conditions, sessile plants are often exposed to varying environmental stresses, such as drought, high salinity and temperature extremes. Drought is a major environmental stress that significantly limits crop production and threatens the sustainable development of agriculture (Zhao and Running, 2010), particularly in arid and semiarid areas. With the exacerbation of global warming, drought stress is occurring more and more frequently in some regions, further threatening crop production and food security. A major effect of drought is the accumulation of reactive oxygen species (ROS) such as superoxide radicals ($O_2^{\cdot-}$), hydrogen peroxide (H_2O_2) and hydroxyl radicals ($\cdot OH$), which cause peroxidation of membrane lipids and damage proteins and nucleic acids (Cruz de Carvalho, 2008). Under water-deficit conditions, excessive ROS combined with closed stomata, smaller leaf area and other metabolic changes, decrease leaf photosynthesis and thereby reduce crop yields (Foyer and Shigeoka, 2011; Gururani et al., 2015; Miyao, 1994; Nishiyama et al., 2011; Tikkanen et al., 2011).

Plants have evolved a series of mechanisms to cope with drought stress, including morphological, physiological, biochemical and molecular adaptations (Anjum et al., 2011; Farooq et al., 2009; Santos et al., 2019). For instance, morphological changes in root system architecture enhance the plant's ability to absorb water by coordinating with cell division, elongation and differentiation in response to water deficit (Dinnyen, 2019; Gupta et al., 2020; Rellán-Alvarez et al., 2016). In general, root system size is positively correlated with drought tolerance. Under water-limited conditions, plant shoot growth is inhibited, but roots have the ability to continue to elongate, resulting in an increased root-to-shoot ratio that promotes survival during water deficit (Yamaguchi and Sharp, 2010). To scavenge the excess ROS generated during drought stress, plants can deploy both enzymatic and non-enzymatic antioxidants. Major antioxidant enzymes include superoxide dismutase (SOD), catalase (CAT) and peroxidase (POD), and flavonoids can function as non-enzymatic antioxidants (Xie et al., 2018).

Increased cuticular wax is another prominent drought response mechanism. Cuticular wax is a mixture of very-long-chain fatty acids and their esters and derivatives, as well as polyketides and terpenoids (Jetter et al., 2006). Forward and reverse genetic approaches have led to the identification of a large number of enzymes, transporters and transcription factors that are involved in the biosynthesis, transport and regulation of cuticular wax. These include CER genes, ATP-BINDING CASSETTE G genes (ABCGs), AP2 transcription factors, MYB94, MYB96 and MYB16 (Lewandowska et al., 2020; Oshima et al., 2013). Wax has a critical role in restricting non-stomatal water loss and reducing epidermal permeability. In response to drought, plants significantly increase their cuticular wax load (Kosma et al., 2009). In addition, a positive correlation between wax amount and drought resistance has been observed. For example, overexpression of *MYB96* or *ECER-IFERUM1* (*CER1*) promotes wax biosynthesis and drought stress tolerance (Bourdenx et al., 2011; Seo et al., 2011). When *MYB94* expression is elevated, plants accumulate more cuticular waxes and lose less water under drought stress (Lee et al., 2016).

The phytohormone auxin is involved in almost all aspects of plant development. Several studies have suggested that auxin biosynthesis is essential for drought tolerance, while others provide evidence that increased drought tolerance is associated with decreased indole-3-acetic acid (IAA) levels (Blakeslee et al., 2019). IAA is the main form of auxin in plants, and auxin homeostasis is regulated by IAA conjugation (Brunoni et al., 2020). GRETCHEN HAGEN3 (GH3) family genes are involved in the conjugation of IAA with amino acids (Mellor et al., 2016; Staswick et al., 2005), forming reversible or irreversible IAA catabolites (Kowalczyk and Sandberg, 2001; Ostin et al., 1998). IAA conjugates with alanine, leucine and phenylalanine function as transient storage compounds and can be hydrolyzed to free IAA, whereas IAA conjugates with aspartate or glutamic acid undergo oxidative metabolism rather than hydrolysis to free IAA (Brunoni et al., 2020; Kowalczyk and Sandberg, 2001; LeClere et al., 2002; Mellor et al., 2016). Previous studies have shown that GH3 family genes play important roles in plant development and stress responses (Singh et al., 2014). GH3.3, GH3.5 and GH3.6 positively and redundantly regulate adventitious root number (Gutierrez et al., 2012; Sorin et al., 2006). Single mutants of *gh3.3*, *gh3.5* or *gh3.6* produce more adventitious roots than the wild-type, whereas *gh3.3-1gh3.5-2gh3.6-1* triple mutants display significantly fewer adventitious roots (Gutierrez et al., 2012). Overexpression of *OsGH3-2* in rice decreases drought tolerance and abscisic acid (ABA) levels (Du et al., 2012), and cotton GH3.5 enhances drought and salt tolerance (Kirungu et al., 2019). In apple (*Malus × domestica*), GH3 family members can be divided into two groups (Yuan et al., 2013).

MsGH3.5 belongs to group II, and inhibits shoot and root growth by modulating the coordination between auxin and cytokinin levels (Zhao et al., 2020). Elevated expression of *MsGH3.5* in the apple rootstock M26 significantly reduces IAA content but increases cytokinin content, resulting in a dwarf phenotype with fewer adventitious roots (Zhao et al., 2020). However, whether *MdGH3* genes play roles in apple drought tolerance is largely unknown.

Previously, we performed an RNA-seq analysis and found that *MsGH3.6* was induced by drought in roots of *Malus sieversii* (Geng et al., 2019). In the present work, we demonstrate that *MdGH3.6* is a negative regulator for apple water-deficit stress tolerance, which is different to the positive roles of cotton GH3.5 and rice GH3.13 in drought tolerance (Kirungu et al., 2019; Zhang et al., 2009).

RESULTS

Elevated expression of *MdGH3.6* reduces the water-deficit stress tolerance of apple

Our previous research showed that *MsGH3.6* was induced by polyethylene glycol (PEG) treatment in apple roots (Figure 1a; Geng et al., 2019). To further investigate *MdGH3.6* (*MDP0000402444*) expression in response to water-deficit stress, we exposed *Malus × domestica* cv. Golden Delicious grafted to *Malus hupehensis* to a short-term water-deficit treatment; leaves were then collected and used for the reverse transcriptase-quantitative polymerase chain reaction (RT-qPCR) assay. *MdGH3.6* expression was repressed shortly after the initiation of water-deficit stress but was increased after 8 days of treatment (Figure 1b), suggesting that it potentially played a role in the water-deficit stress response of apple. Organ-specific expression analysis indicated that *MdGH3.6* was predominantly expressed in roots, followed by stems and flowers; its expression was lowest in leaves (Figure 1c). Subcellular localization analysis suggested that *MdGH3.6* was localized in the plasma membrane, nucleus and cytosol (Figure S1).

To investigate the roles of *MdGH3.6* in the apple water-deficit stress response, we generated transgenic apple plants overexpressing *MdGH3.6* (Figure S2). As shown in Figure S3, the shoot and root lengths of *MdGH3.6* overexpression plants were much shorter than those of non-transgenic GL-3 plants. The shoot biomass of *MdGH3.6* overexpression plants was also less than that of GL-3 plants. After treatment with water-deficit stress, only 52% of *MdGH3.6* overexpression plants survived compared with 70% of GL-3 plants (Figure 2a,b). Water-deficit stress leads to damage of the cell membrane. After water-deficit stress, the electrolyte leakage and malondialdehyde (MDA) content were higher in *MdGH3.6* transgenic plants compared with GL-3 plants (Figure 2c,d). In addition, the photosynthetic capacity of *MdGH3.6* overexpression plants

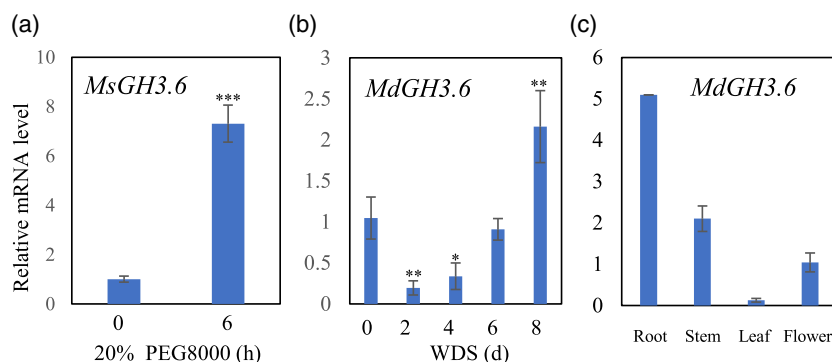


Figure 1. Expression pattern of *MdGH3.6*.

(a) Expression of *MsGH3.6* in response to polyethylene glycol (PEG) treatment in roots of *Malus sieversii*. Hydroponically cultured *M. sieversii* was treated with 20% PEG8000 for 0 or 6 h.

(b) Expression of *MdGH3.6* in response to water-deficit stress. Water was withheld from 2-month-old *Malus × domestica* cv. Golden Delicious for 0, 2, 4, 6 and 8 days, and leaves were collected for RNA extraction.

(c) Expression of *MdGH3.6* in different organs. WDS, water-deficit stress. Error bars indicate SD ($n = 3$). One-way ANOVA (Tukey test) was performed and statistically significant differences were indicated by: * $P < 0.05$; ** $P < 0.01$, *** $P < 0.001$.

was lower than that of GL-3 plants under control and water-deficit stress conditions (Figure S4). Water-deficit stress triggers oxidative stress caused by ROS (Mittler et al., 2004). After water-deficit stress, *MdGH3.6* overexpression plants contained 22–33 $\mu\text{mol g}^{-1}$ H_2O_2 , whereas GL-3 plants had 14 $\mu\text{mol g}^{-1}$ H_2O_2 (Figure 2e). CAT and POD activity was consistently lower in the *MdGH3.6* overexpression plants compared with GL-3 plants (Figure 2f,g), indicating that *MdGH3.6* overexpression plants had lower ROS scavenging ability.

GH3 family genes are involved in the conjugation of IAA with amino acids (Mellor et al., 2016; Staswick et al., 2005; Zhao et al., 2020); we therefore measured the IAA content of *MdGH3.6* overexpression plants under control and water-deficit stress conditions. We found that *MdGH3.6* overexpression plants contained less IAA than GL-3 plants both under control and water-deficit stress conditions, although water-deficit decreased IAA levels in both GL-3 and *MdGH3.6* overexpression plants (Figure S5), a finding consistent with previous research (Zhao et al., 2020).

Knockdown of *MdGH3.6* and its close paralogs increases survival of apple under water-deficit stress

The apple genome encodes two *MdGH3.6* genes: *MdGH3.6* and *MdGH3.6 like* (MDP0000132162). Protein alignment revealed that *MdGH3.6* displayed high sequence similarity with *MdGH3.1* proteins, including MDP0000209432, MDP0000873893, MDP0000872660 and MDP0000666539 (Figure S6). *MdGH3.6 like* and *MdGH3.1* genes also exhibited expression patterns similar to those of *MdGH3.6* under water-deficit stress (Figure S7). To generate transgenic plants with reduced expression of *MdGH3.6*, we transiently knocked down both *MdGH3.6* and *MdGH3.6 like*. However, we found that the expression of *MdGH3.1* genes was also reduced (Figure S8), which might stem from the high

sequence similarity among these genes. To further investigate the function of *MdGH3.6*, we knocked down all these six *MdGH3* genes simultaneously in a GL-3 background using an RNAi approach; the transgenic plants were called *MdGH3* RNAi. As expected, the transgenic RNAi plants had reduced levels of *MdGH3.6*, *MdGH3.6 like*, MDP0000209432, MDP0000873893, MDP0000872660 and MDP0000666539 (Figure S9). In contrast to the shorter roots of *MdGH3.6* overexpression plants, the adventitious root length and number of *MdGH3* RNAi plants were longer and greater than the non-transgenic plants, respectively. The shoot biomass of *MdGH3* RNAi plants was also more than that of GL-3 plants (Figure S10).

We next investigated whether repression of these *MdGH3* genes would affect resistance to water-deficit stress. After a 10-day water-deficit stress treatment and a 7-day recovery, 70–77% of the RNAi plants survived, but only 39% of the GL-3 plants were alive (Figure 3a,b). Further analysis revealed that water-deficit stress resulted in increased plasma membrane damage, as indicated by increased electrolyte leakage and higher MDA content. Ion leakage of GL-3 plants increased by 79% under water-deficit stress, but only a 49–56% increase was observed in RNAi plants (Figure 3c). MDA content increased by 65% in the GL-3 plants after water-deficit stress but by only 52–59% in RNAi plants (Figure 3d). These data indicated that the RNAi plants performed better than the GL-3 plants under short-term water-deficit stress.

H_2O_2 content increased in all plants after short-term water-deficit stress. However, the RNAi plants produced less H_2O_2 under both control and stress conditions. After water-deficit stress, the GL-3 plants contained 8.17 $\mu\text{mol g}^{-1}$ H_2O_2 , whereas the RNAi plants only contained 4.5–6.5 $\mu\text{mol g}^{-1}$ H_2O_2 (Figure 3e). Plants have evolved antioxidant enzymes that scavenge ROS under

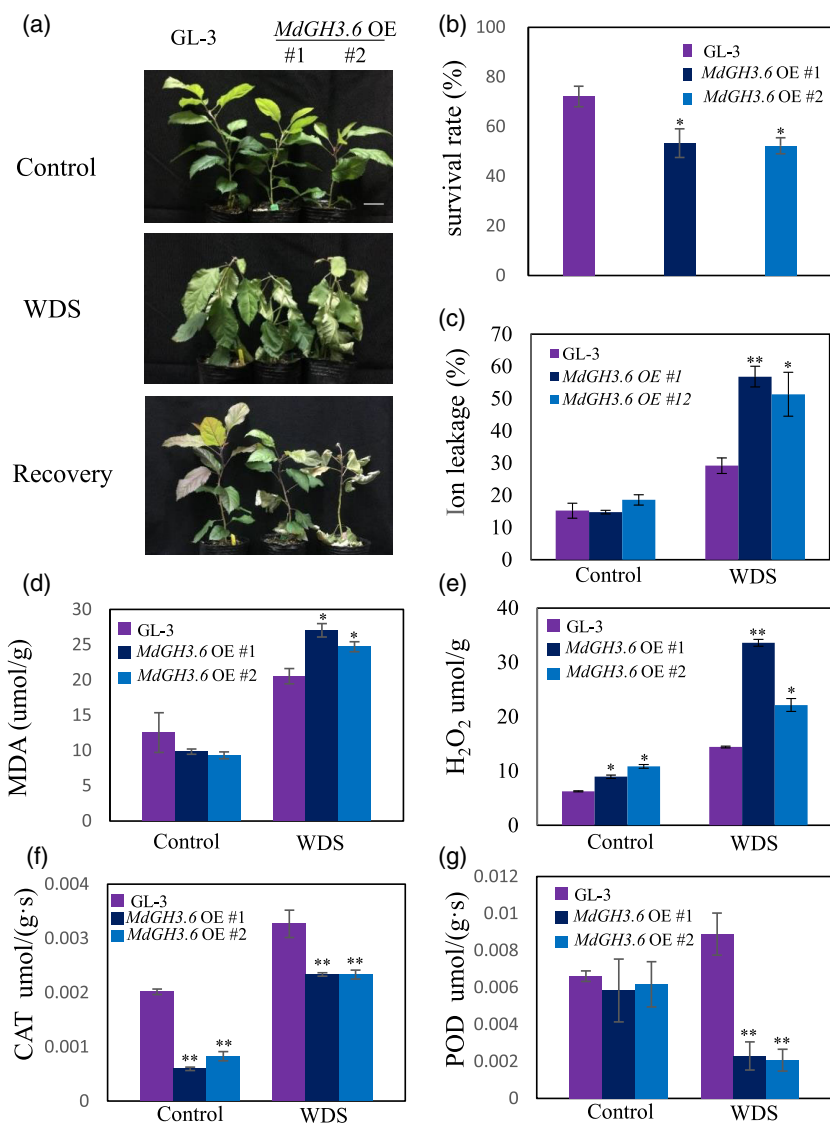


Figure 2. *MdGH3.6* OE plants are more sensitive to water-deficit stress.

(a) Water-deficit stress tolerance of *MdGH3* OE plants. Two-month-old plants were transferred to soil for an additional 2 months, water was then withheld until the soil volumetric water content (VWC) reached about 0%, followed by recovery with watering for 7 days. During the treatment, the GL-3 and *MdGH3.6* OE plants were supplemented with water to maintain the same VWC. Scale bar: 5 cm.

(b) Survival rate of plants shown in (a).

(c) Electrolyte leakage of *MdGH3* OE plants under control and water-deficit stress conditions.

(d) Malondialdehyde (MDA) content of *MdGH3* OE plants under control and water-deficit stress conditions.

(e–g) H_2O_2 content (e), catalase (CAT) activity (f) and peroxidase (POD) activity (g) in *MdGH3* OE plants under control and water-deficit stress. Two-month-old plants were transferred to soil for an additional 2 months, water was then withheld until VWC reached about 10%, and leaves were collected for the measurements. An initial VWC of 50% was required before the treatment. OE, overexpression; WDS, water-deficit stress. Error bars indicate SD [$n = 6$ in (b,c) or 3 in (d–g)]. One-way ANOVA (Tukey test) was performed, and statistically significant differences were indicated by: * $P < 0.05$; ** $P < 0.01$.

water-deficit stress (Cruz de Carvalho, 2008). Here, we found that RNAi plants had higher levels of CAT and POD activity than GL-3 plants under control and water-deficit stress conditions (Figure 3f,g). These results indicated that the RNAi plants accumulated fewer ROS and had higher levels of antioxidant enzymes.

Short-term water-deficit stress also decreased the photosynthetic capacity, intercellular CO_2 concentration and transpiration rate of all plants, although these decreases were less pronounced in *MdGH3* RNAi plants. After 4 days of water-deficit stress, a decrease of only 1–6% in photosynthetic capacity was observed in *MdGH3* RNAi plants; by contrast, a decrease of 23% was observed in GL-3 plants. After 10 days of water-deficit stress, a decrease of 58% in photosynthetic capacity was observed in *MdGH3* RNAi plants, whereas a decrease of 79% was observed in GL-3 plants (Figure S11).

We also measured the IAA content of *MdGH3* RNAi plants under control and water-deficit stress conditions. Under both conditions, *MdGH3* RNAi plants contained more IAA than the GL-3 plants (Figure S12).

Taken together, the above data suggested that *MdGH3.6* is a negative regulator for apple water-deficit stress tolerance, which is different from the positive roles of cotton GH3.5 and rice GH3.13 in water-deficit stress tolerance (Kirungu et al., 2019; Zhang et al., 2009).

Gene expression regulation of *MdGH3.6* under water-deficit stress

The drought response is mediated by a series of drought-responsive genes. To investigate whether *MdGH3.6* mediates the expression of drought-responsive genes, we performed a genome-wide gene expression profiling analysis of GL-3 and RNAi plants under control and water-

Figure 3. *MdGH3* RNAi plants are more tolerant to water-deficit stress.

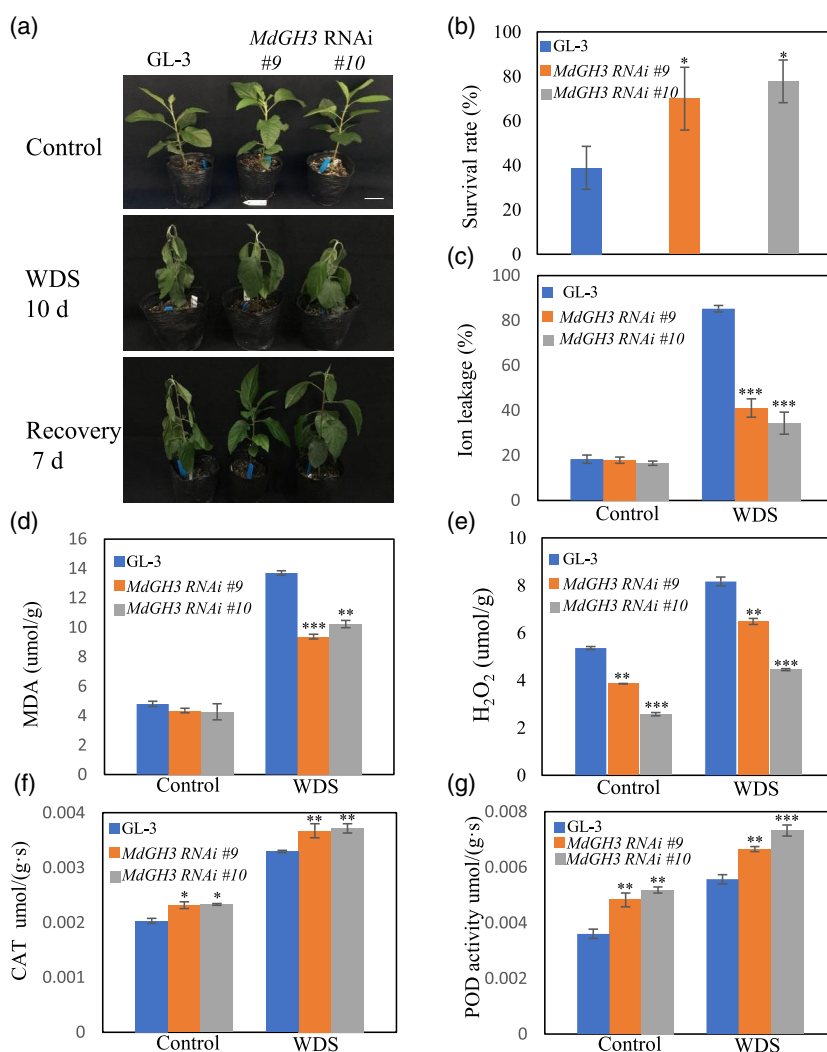
(a) Water-deficit stress tolerance of *MdGH3* RNAi plants. Two-month-old plants were transferred to soil for an additional 1 month, and water was then withheld for 10 days. Scale bar: 3 cm.

(b) Survival rate of plants shown in (a).

(c) Electrolyte leakage of *MdGH3* RNAi plants under control and water-deficit stress conditions.

(d) Malondialdehyde (MDA) content of *MdGH3* RNAi plants under control and water-deficit stress conditions. Plants were exposed to water-deficit stress for 10 days, and leaves were collected for an ion leakage assay and MDA content measurement.

(e–g) H_2O_2 content (e), catalase (CAT) activity (f) and peroxidase (POD) activity (g) in *MdGH3* RNAi plants under control and water-deficit stress. Plants were exposed to water-deficit stress for 0 and 10 days, and leaves were collected for the measurements. WDS, water-deficit stress. Error bars indicate SD [$n = 6$ in (b,c) or 3 in (d–g)]. One-way ANOVA (Tukey test) was performed, and statistically significant differences were indicated by: * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.



deficit stress conditions (Datasets S1–S6). After 10 days of water-deficit stress, 5684 genes were upregulated in the GL-3 plants, and 5189 genes were downregulated (Datasets S1 and S2). Under control conditions, 5541 genes were upregulated and 3658 genes were downregulated in the *MdGH3* RNAi plants compared with the GL-3 plants (Datasets S3 and S4). Under water-deficit stress conditions, 1099 genes were upregulated and 497 genes were downregulated in the *MdGH3* RNAi plants compared with the GL-3 plants ($P < 0.05$, 2.0-fold cutoff; Datasets S5 and S6). Among the 1099 genes upregulated in RNAi plants under water-deficit stress, 114 (10%) were induced by water-deficit stress. Among the 497 downregulated genes, 56 (11%) were water-deficit stress -repressed genes (Figures 4a,b and S13; Datasets S5 and S6). The genes up- and downregulated in RNAi plants may contribute to water-deficit stress resistance. Among the differentially expressed genes (DEGs) in the RNAi plants under water-deficit stress were genes related to ROS scavenging, cuticular wax biosynthesis, auxin transport and drought stress

response (Datasets S3 and S4). We selected eight genes to verify the results of the transcriptome analysis: *MdPIN1*, *MdPIN5*, *Peroxidase p7-like*, *Peroxidase 4-like*, *MdCHS*, *MdFSD2*, *MdERF1* and *MdWRKY70*. These genes are involved in drought tolerance, auxin transport, oxidation–reduction, chalcone biosynthesis and ethylene signaling (Bolwell et al., 2002; Chen et al., 2017; Cheng et al., 2017; Fernández-Pérez et al., 2015; Mravec et al., 2009; Myouga et al., 2008; Wang et al., 2018; Willige et al., 2011). Expression of all genes except for *MdWRKY70* was upregulated after water-deficit stress in RNAi plants, consistent with the RNA-seq results (Figure 4c). We also examined the expression of these genes in the *MdGH3.6* overexpression plants. The results revealed that *MdGH3.6* positively regulated *MdWRKY70* expression, but negatively regulated the expression of seven other genes under water-deficit stress conditions (Figure 4d).

We then performed Gene Ontology (GO) analysis of genes up- and downregulated in RNAi plants under water-deficit stress. In terms of biological processes, upregulated

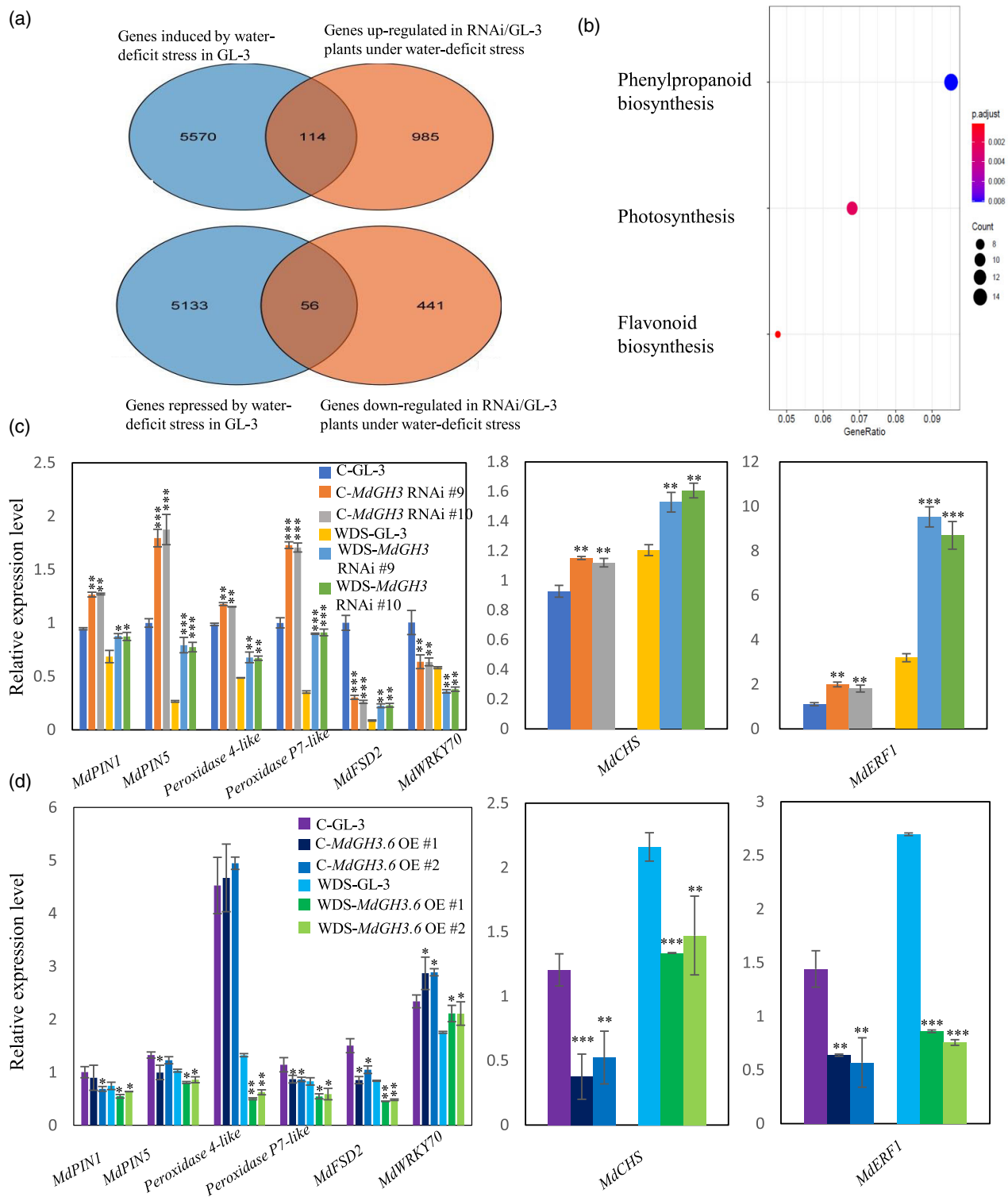


Figure 4. RNA-seq analysis of *MdGH3.6* transgenic plants in response to water-deficit stress.

(a) Venn diagrams showing the numbers of genes induced by water-deficit stress, and genes upregulated or downregulated in *MdGH3* RNAi plants under water-deficit stress.

(b) KEGG enrichment of genes upregulated in *MdGH3* RNAi plants under water-deficit stress.

(c,d) Verification of transcriptome analysis by reverse transcriptase-quantitative polymerase chain reaction (RT-qPCR). C, control; WDS, water-deficit stress. Error bars indicate SD ($n = 3$). One-way ANOVA (Tukey test) was performed, and statistically significant differences were indicated by: * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

DEGs were enriched in response to stimuli, response to chemicals, oxidation–reduction process and flavonoid biosynthesis (Figure S14). DEGs downregulated in RNAi plants under water-deficit stress were significantly enriched in metabolic process (Figure S15). KEGG analysis revealed that DEGs upregulated in RNAi plants under water-deficit stress were enriched in phenylpropanoid biosynthesis, photosynthesis and flavonoid synthesis (Figure 4b). By contrast, DEGs downregulated in RNAi plants under water-deficit stress were enriched in pentose and glucuronate interconversion (Figure S16).

MdGH3.6 negatively regulates leaf cuticular wax and phenylpropanoid and flavonoid biosynthetic metabolites under water-deficit stress

Among the DEGs, several genes were involved in cuticular wax biosynthesis and transport, including *MdMYB16*, *MdACBP6*, *MdNAT*(Acyl-CoA *N*-acyltransferases superfamily protein) and *MdABCG15* (Fang et al., 2017; Oshima et al., 2013; Qin et al., 2013; Xia et al., 2012; Dataset S3). RT-qPCR further confirmed that the expression of these genes was upregulated in *MdGH3* RNAi plants but downregulated in *MdGH3.6* overexpression plants under water-deficit stress (Figure 5a,b). We also found that *MdGH3.6* negatively regulated other cuticular wax biosynthesis-related genes, including *MdACBP1*, *MdKCS1*, *MdKCR1*, *MdHDG2* (Lee et al., 2016; Li et al., 2020; Xue et al., 2014; Figure S17). To investigate the association between the expression of these genes and cuticular wax content, we measured the content of 10 leaf cuticular wax monomers in GL-3 and *MdGH3* RNAi plants under control and long-term water-deficit stress conditions. As shown in Figure 5 (c), C29 and C31 alkanes and C26 alkanol were the predominant wax monomers in apple leaves under water-deficit stress, accounting for 72% of all monomers. Under control conditions, there were no significant differences in the contents of six leaf cuticular wax components between RNAi plants and GL-3 plants. The four exceptions were heptacosane (C27 alkane), hentriacontane (C31 alkanes), 1-hexacosanol (C26 alkanol) and octacosanol (C28 alkanol), which were higher in RNAi plants (Figure 5c). Water-deficit stress increased the contents of most cuticular wax monomers in all plants, including myristic acid (C14 fatty acid), palmitic acid (C16 fatty acid), heptacosane (C27 alkane), nonacosane (C29 alkane), 1-hexacosanol (C26 alkanol), hentriacontane (C31 alkane) and octacosanol (C28 alkanol). After long-term water-deficit stress, the contents of nine out of 10 wax monomers differed significantly between GL-3 and RNAi plants. The exception was hexacosane (C26 alkane), whose content was similar between the two genotypes. The contents of heneicosane, myristic acid, palmitic acid, octadecanoic acid, heptacosane, nonacosane, 1-hexacosanol, hentriacontane and octacosanol were higher in RNAi plants than in GL-3 plants (Figure 5c).

Consistent with their higher leaf cuticular wax content, RNAi plants under water-deficit stress also had lower chlorophyll exudation rates (Figure S18). We then examined the cuticular wax content in *MdGH3.6* overexpression plants under control and short-term water-deficit stress conditions. C29 and C31 alkanes and C26 alkanol were less abundant in *MdGH3.6* overexpression plants under water-deficit stress (Figure S19). Leaf cuticular waxes, particularly C29, are positively associated with drought tolerance (Panikashvili et al., 2007), and the wax content of *MdGH3* RNAi transgenic plants probably contributed to their water-deficit stress resistance. Taken together, the above data suggest that the cuticular wax content is negatively regulated by *MdGH3.6*.

Metabolites in the phenylpropanoid and flavonoid biosynthetic pathways are important for plant abiotic and biotic stress tolerance, as they are capable of scavenging ROS (Geng et al., 2020). KEGG analysis of upregulated DEGs revealed enrichment in genes associated with phenylpropanoid and flavonoid biosynthesis in RNAi plants under water-deficit stress (Figure 4b). We therefore asked whether levels of metabolites involved in phenylpropanoid and flavonoid biosynthesis were altered in RNAi plants under long-term water-deficit stress. Using high-performance liquid chromatography-mass spectrometry (HPLC-MS/MS), we measured the levels of 17 metabolites in the phenylpropanoid and flavonoid biosynthetic pathways. Twelve metabolites were increased in RNAi plants after water-deficit stress: chlorogenic acid, naringin, prunin, catechin, procyanidinB2, procyanidinB1, quercetin, p-coumaric acid, caffeic acid, quercetin 3-O-glucoside, phlorizin and hyperin (Figure 6). Increased phenylpropanoid and flavonoid metabolites in RNAi plants may have contributed to their increased ROS scavenging ability during water-deficit stress. By contrast, contents of these metabolites were lower in *MdGH3.6* overexpression plants than in GL-3 plants under water-deficit stress, suggesting the negative regulation of these metabolites by *MdGH3.6* under water-deficit stress (Figure 6).

MdGH3.6 is a direct target of MdMYB94 and is negatively regulated by MdMYB94

To further understand the molecular mechanisms of *MdGH3.6* in response to water-deficit stress, we analyzed the promoter of *MdGH3.6*, and identified one MYB binding site (Motif2) and one binding site of AtMYB94 and AtMYB96 (Motif 1; Figure 7a; Lee et al., 2016). Considering the critical roles of MYB genes in cuticular wax and drought stress tolerance, we performed dual-luciferase assays. MdMYB94 (MDP0000144744), rather than MdMYB96 (MDP0000149535), could bind to the promoter of *MdGH3.6* (Figures 7b and S20). We further conducted a DAP-qPCR analysis and found that MdMYB94 could bind to the two motifs of the *MdGH3.6* promoter, and the

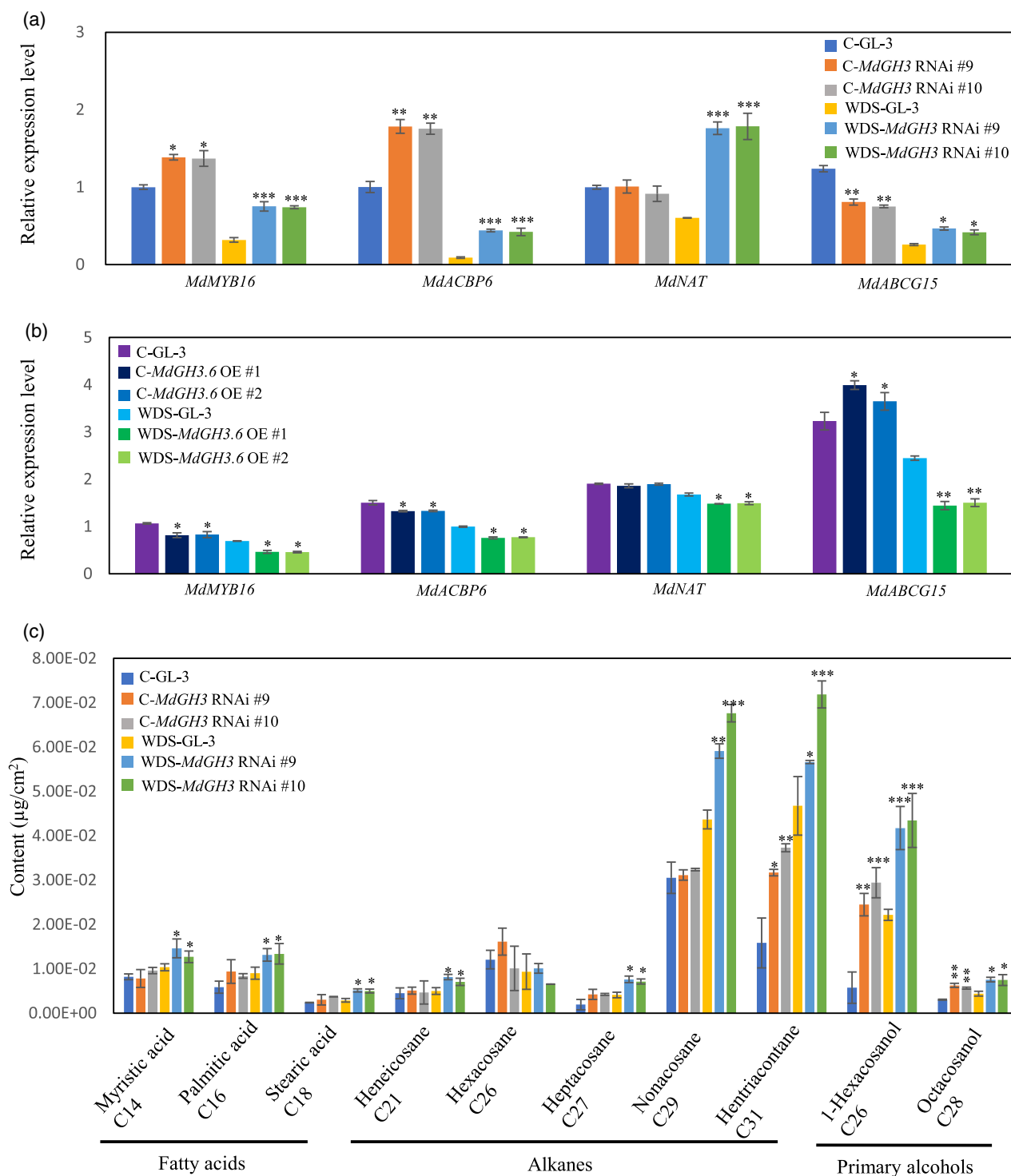


Figure 5. Cuticular wax content of GL-3 and *MdGH3* RNAi plants under long-term water-deficit stress.

(a,b) Expression of wax biosynthesis-related genes in GL-3, *MdGH3* RNAi or *MdGH3.6* OE plants in response to water-deficit stress. OE, overexpression. (c) Cuticular wax content in GL-3 and *MdGH3* RNAi plants after long-term water-deficit stress conditions. Plants were exposed to water-deficit stress for 60 days. Error bars indicate SD ($n = 3$). C, control; WDS, water-deficit stress. Cuticular wax content was measured by gas chromatography-mass spectrometry (GC-MS). One-way ANOVA (Tukey test) was performed, and statistically significant differences were indicated by: * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

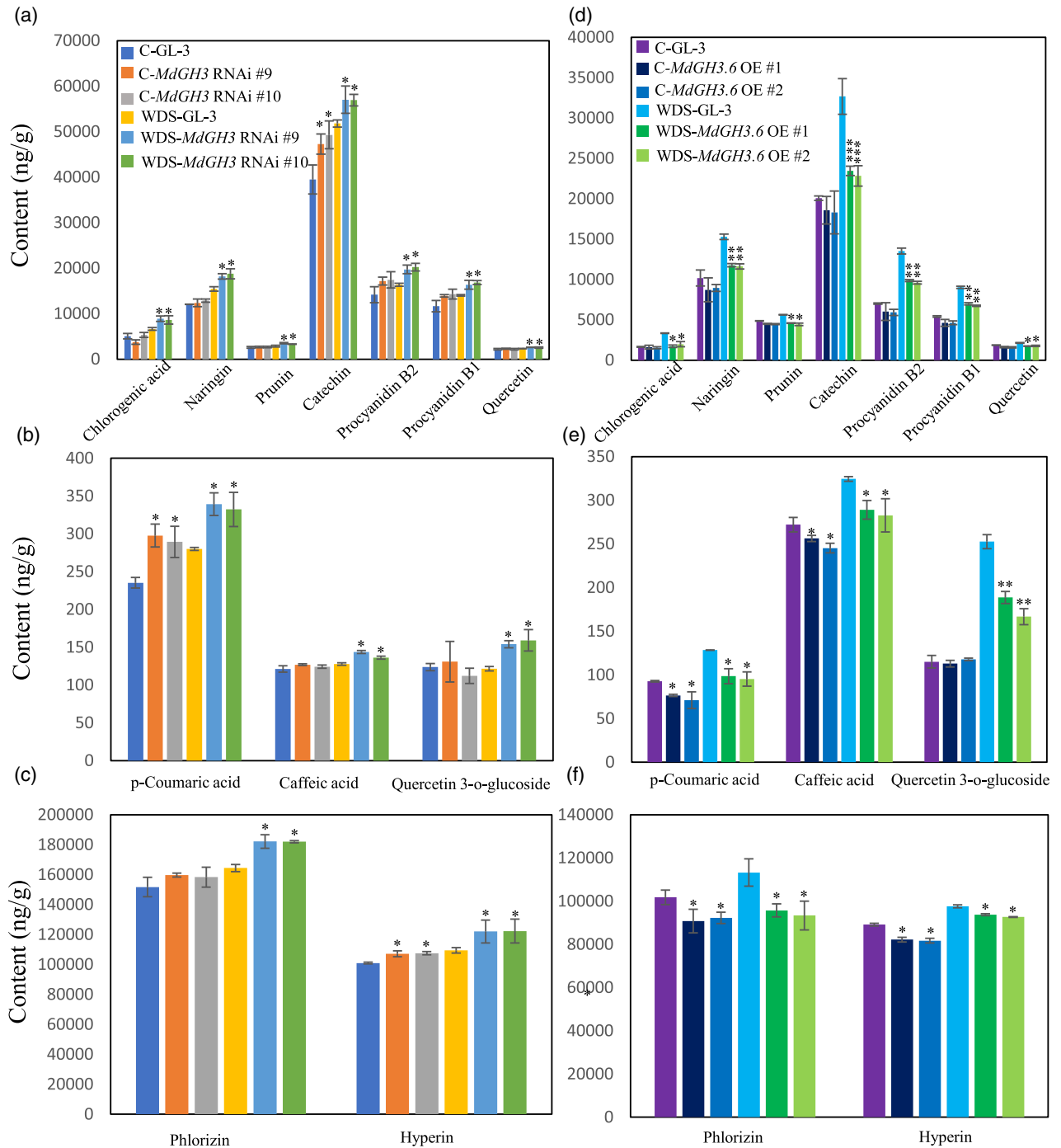


Figure 6. Content of metabolites involved in the phenylpropanoid and flavonoid pathway in GL-3, *MdGH3* RNAi plants and *MdGH3.6* OE plants under water-deficit stress.

Two-month-old *MdGH3.6* OE plants were transferred to soil for an additional 2 months, water was then withheld until volumetric water content (VWC) reached about 10%, and leaves were collected for the measurements. For the *MdGH3* RNAi plants, a long-term water-deficit stress lasting for 60 days was performed. Metabolites were measured by LC-MS/MS. C, control; WDS, water-deficit stress. Error bars indicate SD ($n = 5$). One-way ANOVA (Tukey test) was performed, and statistically significant differences were indicated by: * $P < 0.05$; ** $P < 0.01$.

binding to motif 1 was stronger (Figure 7c). The chromatin immunoprecipitation (ChIP)-qPCR analysis and electrophoretic mobility shift assay (EMSA) assay further

confirmed that MdMYB94 could bind to motif 1 of the *MdGH3.6* promoter (Figure 7d,e). In addition, MdMYB94 negatively regulated the *MdGH3.6* promoter under control

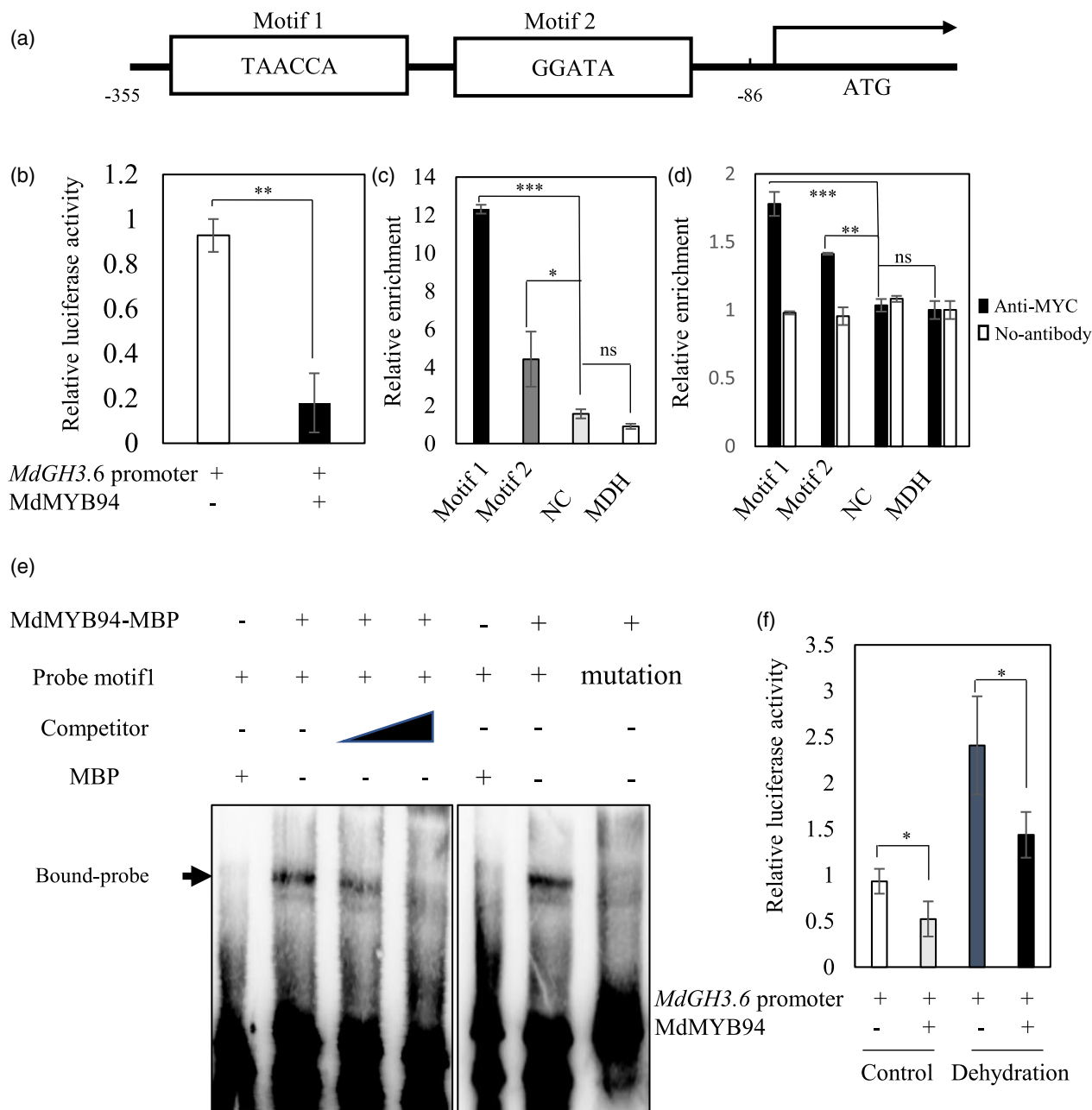


Figure 7. *MdGH3.6* is a direct target of MdMYB94.

(a) Schematic diagram showing the two binding motifs in promoter of *MdGH3.6*.

(b) MdMYB94 can bind to *MdGH3.6* promoter by a dual-luciferase assay.

(c–e) Direct binding of MdMYB94 to the motif 1 in *MdGH3.6* promoter using DAP-quantitative polymerase chain reaction (qPCR) (c), chromatin immunoprecipitation (ChIP)-qPCR (d) or electrophoretic mobility shift assay (EMSA) analysis (e). NC, negative control; ns, no significance.

(f) MdMYB94 can repress the expression of *MdGH3.6* under dehydration conditions. Error bars indicate SD [$n = 3$ in (c,d) or 6 in (b,f)]. One-way ANOVA (Tukey test) was performed, and statistically significant differences were indicated by: * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

and dehydration conditions (Figure 7f). These results suggest that *MdGH3.6* is a direct target of MdMYB94.

MdMYB94 positively regulates apple water-deficit stress tolerance and leaf cuticular wax biosynthesis

The homolog of MdMYB94 in Arabidopsis is significantly induced by drought stress, and positively associated with

cuticular wax and drought tolerance (Lee et al., 2016). We also found that *MdMYB94* was induced by water-deficit stress (Figure S21). To investigate the roles of MdMYB94 in the apple water-deficit stress response, we generated transgenic apple calli overexpressing *MdMYB94* (Figure S22). After treatment with 20% PEG8000, the relative fresh weight of the transgenic calli was higher than that of the wild-type

(Figure S22). This result indicated that MdMYB94 may positively regulate apple water-deficit stress. To further verify that MdMYB94 can positively regulate water-deficit stress, we also generated transgenic apple plants overexpressing MdMYB94 (Figure S23). After treatment with water-deficit stress, 83–86% of the MdMYB94 OE plants survived, but only 33% of the GL-3 plants were alive (Figure 8a,b). Ion leakage of GL-3 plants increased to 72% after water-deficit stress, but only a level of 29–32% was observed in MdMYB94 OE plants (Figure 8c). MDA content increased by 56% in the GL-3 plants after water-deficit stress but only 34–36% in MdMYB94OE plants (Figure 8d). These data indicated that the MdMYB94 OE plants perform better than the GL-3 plants under short-term water-deficit stress. We also found that H₂O₂ content increased in all plants after

short-term water-deficit stress. However, the MdMYB94 OE plants produced less H₂O₂ under both control and stress conditions (Figure 8e). Consistently, we found that MdMYB94 OE plants had higher levels of CAT and POD activity than GL-3 plants under control and water-deficit stress conditions (Figure 8f,g). These results indicated that the MdMYB94 OE plants accumulated fewer ROS and had higher levels of antioxidant enzymatic activities. Taken together, the above data suggested that MdMYB94 is a positive regulator for water-deficit stress tolerance of apple.

To investigate the roles of MdMYB94 in wax biosynthesis of apple leaves, we firstly detected the chlorophyll release of GL-3 and MdMYB94 OE plants. We found that the chlorophyll release of transgenic plants was lower than that of GL-3 plants (Figure 9a). This result indicated that

Figure 8. MdMYB94 OE plants are more tolerant to water-deficit stress.

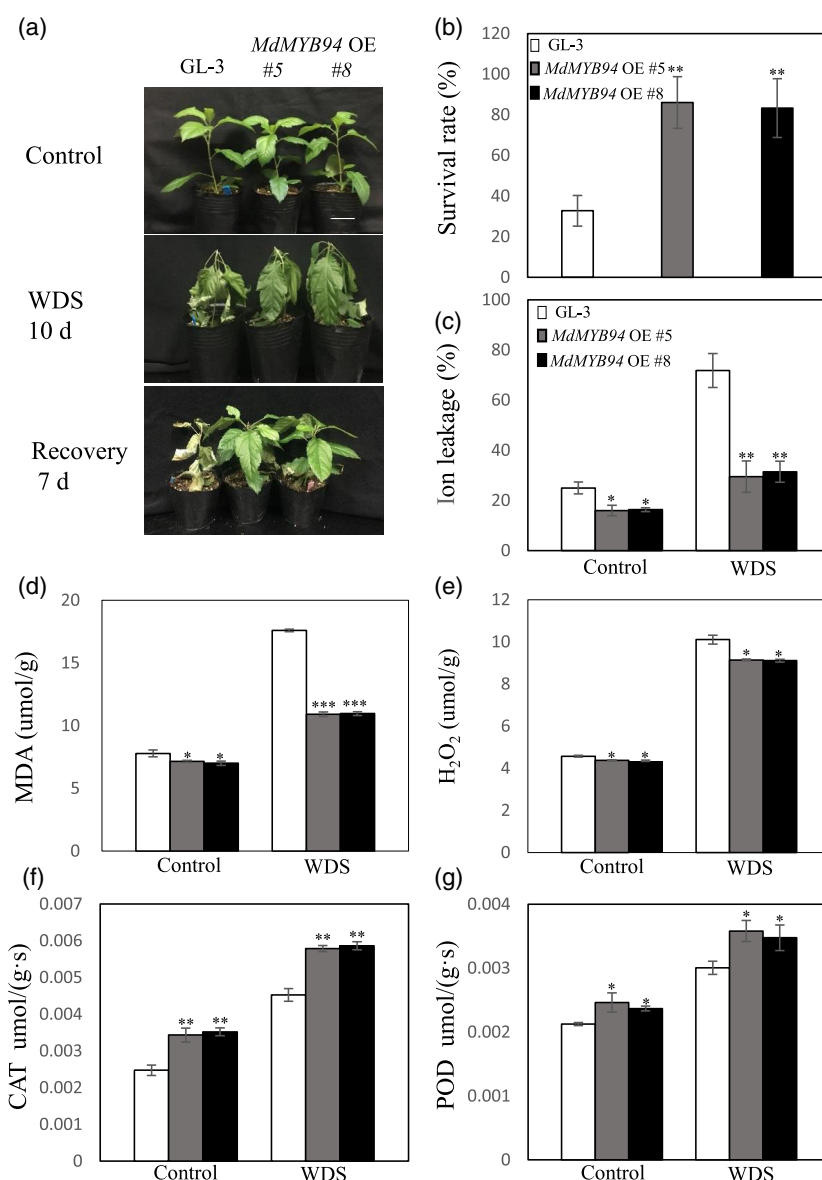
(a) Water-deficit stress tolerance of MdMYB94 OE plants. Two-month-old plants were transferred to soil for an additional 2 months, and water was then withheld for 10 days. Scale bar: 4 cm.

(b) Survival rate of plants shown in (a).

(c) Electrolyte leakage of MdMYB94 OE plants under control and water-deficit conditions.

(d) Malondialdehyde (MDA) content of MdMYB94 OE plants under control and water-deficit conditions. Plants were exposed to water-deficit stress for 10 days, and leaves were collected for an ion leakage assay and MDA content measurement.

(e–g) H₂O₂ content (e), catalase (CAT) activity (f) and peroxidase (POD) activity (g) in MdMYB94 OE plants under control and water-deficit stress. Plants were exposed to water-deficit stress for 0 and 10 days, and leaves were collected for the measurements. WDS, water-deficit stress. Error bars indicate SD [*n* = 6 in (b,c) or 3 in (d–g)]. One-way ANOVA (Tukey test) was performed, and statistically significant differences were indicated by: **P* < 0.05; ***P* < 0.01; ****P* < 0.001.



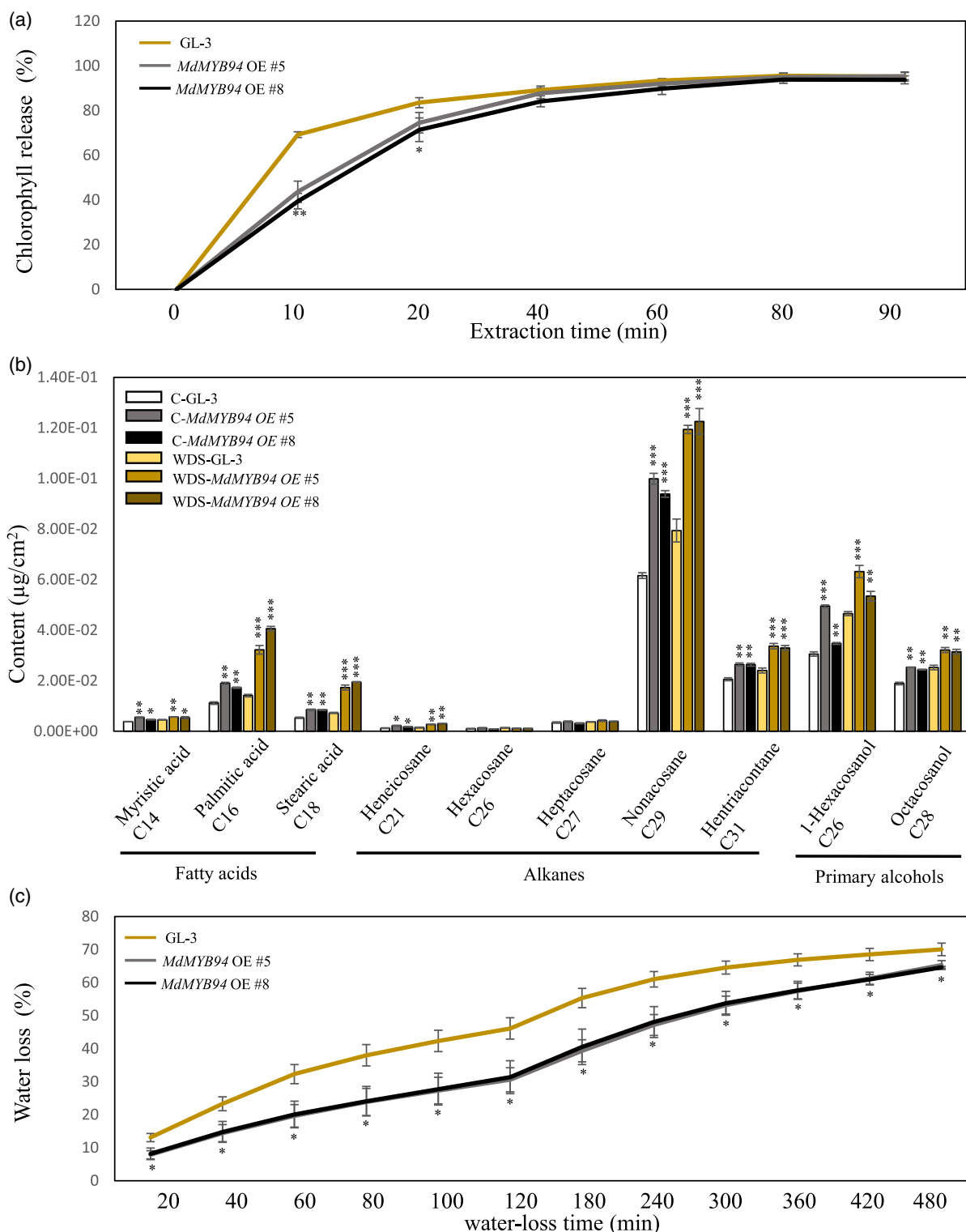


Figure 9. MdMYB94 positively regulate the cuticular wax biosynthesis of apple leaves.

(a) Chlorophyll release from GL-3 and MdMYB94 OE plants. Leaves of 2-month-old plants were collected for chlorophyll extraction.

(b) Cuticular wax content in GL-3 and MdMYB94 OE plants after water-deficit stress conditions. Two-month-old MdMYB94 OE plants were transferred to soil for an additional 2 months, and water was then withheld for 10 days, and leaves were collected for the measurements. Error bars indicate SD ($n = 3$). C, control; WDS, water-deficit stress. Cuticular wax content was measured by gas chromatography-mass spectrometry (GC-MS).

(c) Water loss from GL-3 and MdMYB94 OE plants. Leaves of 2-month-old plants were collected for water loss. Error bars indicate SD [$n = 6$ in (a,c)]. One-way ANOVA (Tukey test) was performed, and statistically significant differences were indicated by: * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

MdMYB94 may positively regulate wax biosynthesis of apple leaves. To further verify that MdMYB94 is involved in wax biosynthesis of apple leaves, we measured the content of 10 leaf cuticular wax monomers in GL-3 and *MdMYB94* OE plants under control and water-deficit stress conditions. As shown in Figure 9, most cuticular wax monomers were more abundant in *MdMYB94* OE plants under control and water-deficit stress (Figure 9b), including myristic acid (C14 fatty acid), palmitic acid (C16 fatty acid), stearic acid (C18 fatty acid), nonacosane (C29 alkane), 1-hexacosanol (C26 alkanol), hentriacontane (C31 alkane) and octacosanol (C28 alkanol). In addition, we found that *MdMYB94* plants had lower water loss than GL-3 plants (Figure 9c), which might contribute to their water-deficit stress resistance. Taken together, the above data suggested that MdMYB94 can positively regulate wax biosynthesis of apple leaves.

DISCUSSION

In this study, we identified a negative regulator of water-deficit stress tolerance, MdGH3.6, in apple. Overexpressing *MdGH3.6* in apple reduced water-deficit stress tolerance; however, the reduced expression of *MdGH3.6* and its close paralogs increased water-deficit stress tolerance (Figures 1 and S7). *MdMYB94* is activated under water-deficit stress, which then directly reduces the expression of *MdGH3.6*. MdGH3.6 negatively regulates cuticular wax content, root length, antioxidant enzyme activity and secondary metabolites involved in the phenylpropanoid and flavonoid pathway. The latter two can scavenge excessive ROS under water-deficit stress (Figure 10). The negative regulation of wax content by MdGH3.6 may be achieved by negatively regulating *MdMYB94* expression (Figure 8).

Previous research has shown that *GH3* genes play diverse roles in different species under drought stress. Cotton *GH3.5* is a positive regulator of drought tolerance. Silencing *GH3.5* in cotton reduces tolerance to drought stress (Kirungu et al., 2019). The silenced plants had lower chlorophyll content, relative leaf water content and SOD activity, but higher MDA content and ion leakage level under drought conditions (Kirungu et al., 2019). However, *OsGH3-2* is a negative regulator of drought tolerance (Du et al., 2012). Overexpressing *OsGH3-2* in rice reduces the contents of ABA and carotene, but increases water loss and stomatal aperture (Du et al., 2012). The opposite roles of *GH3* genes in different species indicate that *GH3* has diverse functions across plant species; this was further supported by our observations in apple. Interestingly, rice *GH3.13* (TLD1) is a positive regulator of drought tolerance (Zhang et al., 2009). Activation of *GH3.13* enhances drought survival and the expression of *LEAs*, but loss-of-function mutants of *GH3.13* show phenotypes similar to the wild-type under drought stress (Zhang et al., 2009).

This observation indicates that *GH3* genes might play different roles, even in the same species.

GH3 genes are involved in auxin homeostasis by conjugating excess IAA to amino acids. The apple rootstock M26 expressing *MsGH3.5* has less free IAA and fewer adventitious roots (Zhao et al., 2020). Overexpressing *OsGH3-2* in rice causes morphological aberrations including dwarfism, fewer root hairs and fewer crown roots (Du et al., 2012). Gain-of-function rice mutant *tlds-D* is characterized by dwarfism, and has increased tiller number and leaf angles (Zhang et al., 2009). In our study, *GH3.6* overexpression plants had less IAA and fewer adventitious roots, whereas *GH3* RNAi plants contained more IAA and adventitious roots (Figures S3 and S10), a finding consistent with previous research (Zhao et al., 2020). By contrast, Arabidopsis *GH3.3*, *GH3.5* and *GH3.6* are positive regulators of adventitious rooting, and the *gh3.3gh3.5gh3.6* triple mutant shows reduced adventitious root number, despite the fact that the *GH3.6* sequence is highly similar to MdGH3.6 (Gutierrez et al., 2012; Figure S21). The opposite effects of these *GH3* genes on adventitious root development further indicate that *GH3* genes from different plant species may have different functions despite their sequence similarity. It is also possible that there may be different mechanisms in root development in various species. For example, other regulatory genes may be involved in root development in apple and Arabidopsis. However, additional studies are required to investigate this hypothesis.

Auxin is a well-known regulator of plant growth and development that controls root patterning, apical dominance and tropisms. Previous studies have shown that auxin can promote water-deficit tolerance by enhancing ABA signal transduction, increasing ROS scavenging ability and modulating plant growth (Bielach et al., 2017; Naser and Shani, 2016; Shi et al., 2014). Activation of Arabidopsis

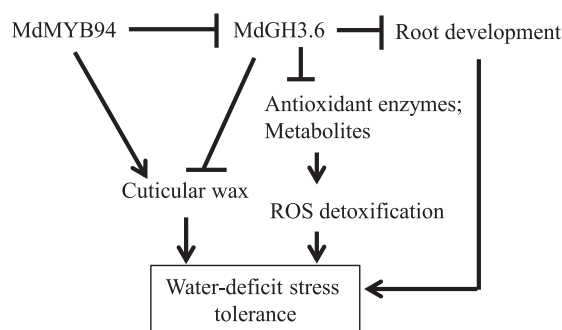


Figure 10. A working model for MdGH3.6 in response to water-deficit stress.

Under water-deficit stress, *MdMYB94* is activated, which then directly reduces expression of *MdGH3.6*. MdGH3.6 negatively regulates cuticular wax content, root length, antioxidant enzyme activity and secondary metabolites involved in the phenylpropanoid and flavonoid pathways. The latter two can scavenge excessive reactive oxygen species (ROS) under water-deficit stress.

YUCCA6 (*YUC6*) in potato or poplar improved drought stress resistance, which was associated with auxin-overproduction morphological phenotypes as well as reduced ROS levels (Ke et al., 2015; Kim et al., 2013). The gain-of-function *yuc7-1D* mutant exhibited increased drought tolerance, along with enhanced apical dominance, superior root growth and upregulated expression of *RESPONSIVE TO DESICCATION 29A* (*RD29A*; Lee et al., 2012). However, an independent study showed that Arabidopsis plants that overexpressed *YUC6* displayed improved drought stress tolerance independent of IAA (Cha et al., 2015). Here, we found that water-deficit stress-sensitive *MdGH3.6* overexpression plants contained less IAA, but water-deficit stress-resistant *MdGH3* RNAi plants contained more IAA (Figures S5 and S12). However, the direct link between auxin and drought stress tolerance remains unclear. Whether IAA plays a key role in determining the drought tolerance of transgenic *MdGH3* RNAi plants and an auxin-independent pathway existed in regulating drought response of apple requires further investigation.

Under drought stress, plants quickly accumulate ROS as an initial defense response (Qi et al., 2018; You and Chan, 2015). ROS can act as signaling molecules, triggering downstream responses and regulating plant growth, the stress response, programmed cell death and hormonal signaling. However, excess ROS are toxic and can cause the oxidative destruction of cells (Mittler et al., 2004). Therefore, an efficient ROS scavenging system is important for plant survival under drought stress. Plants have evolved two strategies to detoxify ROS: the production of enzymatic and non-enzymatic antioxidants. The major enzymes involved in ROS scavenging include SOD, POD, CAT, glutathione peroxidase, glutathione (GSH) and peroxiredoxin (Choudhury et al., 2017). ROS detoxification can also be achieved by some metabolites including ascorbate, GSH, carotenoids, tocopherols and flavonoids (Xie et al., 2018). We found that H_2O_2 accumulation was lower in *MdGH3* RNAi plants under water-deficit stress (Figure 3e), indicating that these plants had a more powerful antioxidant system. Levels of CAT, POD and flavonoids were also consistently and significantly higher in *MdGH3* RNAi plants under water-deficit stress (Figures 3 and 6). In contrast, the H_2O_2 level was higher in *MdGH3.6* overexpression plants under water-deficit stress. Accordingly, levels of CAT, POD and flavonoids were lower in *MdGH3.6* overexpression plants in response to water-deficit stress (Figures 2 and 6). Based on the RNA-seq analysis, we also identified the elevated expression of POD genes in *MdGH3* RNAi plants under water-deficit stress (Figure 4; Dataset S3), providing further evidence that *MdGH3.6* negatively regulated ROS scavengers, including antioxidant enzymes and non-enzymatic metabolites.

Previous studies have shown that cuticular wax is associated with improved drought resistance (Guo et al., 2016;

Meng et al., 2019; Patwari et al., 2019; Xue et al., 2017; Zhu and Xiong, 2013). We found that *GH3* RNAi plants had higher levels of cuticular wax under water-deficit stress compared with *GH3.6* overexpression plants (Figures 5c and S19). More importantly, the wax composition was altered in *GH3.6* transgenic plants after water-deficit stress compared with GL-3 plants. *MdGH3* RNAi plants under water-deficit stress had significantly higher levels of C29 alkane compared with GL-3 plants, consistent with a previous study indicating that lower C29 alkane levels were associated with reduced drought tolerance (Panikashvili et al., 2007). In addition, C31 alkanes were present in amounts similar to C29 alkane, followed by C26 alcohol (Figure 5c). In contrast, *MdGH3.6* overexpression plants had lower levels of C29 and C31 alkanes and C26 alcohol under water-deficit stress (Figure S19), implying that *GH3.6* negatively regulated cuticular wax content under water-deficit stress. We also noticed that in *GH3* RNAi plants, the main cuticular waxes were C29 and C31 alkanes and C26 alcohol; however, in the *GH3.6* overexpression plants under water-deficit stress, the main cuticular waxes were C16 and C18 fatty acids, in addition to C29 and C31 alkanes and C26 alcohol. These differences might stem from the different developmental stages of the plants used for water-deficit stress treatment and the different duration of water-deficit stress treatment.

In sum, we characterized *MdGH3.6* in the response of apple to water-deficit stress. We found that *MdGH3.6* was a direct target of MdMYB94 and a negative regulator of water-deficit stress tolerance in apple. The negative regulation of water-deficit stress tolerance may stem from the combined roles of *MdGH3.6* in the negative regulation of ROS detoxification and cuticular wax (Figure 10).

EXPERIMENTAL PROCEDURES

Plant materials and growth conditions

For *Agrobacterium*-mediated genetic transformation, 1-month-old tissue-cultured GL-3 (selected from progenies of *Malus × domestica* 'Royal Gala') plants were used as described previously (Dai et al., 2013).

For gene cloning and expression in response to water-deficit stress, 'Golden Delicious' (*Malus × domestica*) apple trees grown in a greenhouse were treated with water-deficit stress for 0, 2, 4, 6 and 8 days, and mature leaves were collected for RNA extraction.

Short-term water-deficit stress treatment

For short-term water-deficit stress treatment, GL-3 and transgenic plants were rooted in 1/2 MS medium (2.215 g L⁻¹ MS salts, 20 g L⁻¹ sucrose and 7 g L⁻¹ agar, pH 5.8) supplemented with 0.5 mg L⁻¹ indole-3-butyric acid and 0.5 mg L⁻¹ IAA under dark conditions for 3 days, and then held under long-day conditions (14 h light/10 h dark) for an additional 2 months. Plants were then transplanted into soil and grown in a light growth chamber with 60% humidity under long-day conditions. After 1 or 2 months, water-deficit stress treatment was carried out by withholding

water for 10 days for GL-3, *MdGH3* RNAi and *MdMYB94* OE plants. Survival rate was counted after re-watering for 7 days. For GL-3 and the *MdGH3.6* overexpression plants, water was withheld until the soil volumetric water content (VWC) reached 0%, followed by recovery with watering for 7 days, and survival rate was then counted. The GL-3 and *MdGH3.6* OE plants were supplemented with water to maintain the same VWC during the water-deficit stress treatment. An initial VWC of 50% was required before the treatment. VWC was measured by a soil moisture meter TDR350 (Spectrum Technologies, Aurora, IL, USA).

To measure ion leakage, MDA and H_2O_2 content, CAT and POD activity, gene expression and photosynthetic parameters, the GL-3 and *MdGH3* RNAi plant leaves were collected after water was withheld for 10 days. Photosynthetic rate, stomatal conductance and evaporation rate were monitored with a LI-6400XT portable photosynthesis system (LI-COR, Lincoln, NE, USA). Data were collected from the 4–6th leaves from the base of the stems.

To measure ion leakage, MDA and H_2O_2 content, CAT and POD activity, gene expression, cuticular wax and metabolite content, and photosynthetic parameters, the GL-3 and *MdGH3.6* OE plant leaves were collected when VWC reached 10% during treatment. Plant height and root length were measured by a meter ruler.

Long-term water-deficit stress treatment

A long-term water-deficit stress was performed to measure cuticular wax and metabolite content of GL-3 and *MdGH3* RNAi plants as described previously (Geng et al., 2018). Thirty-six tissue-cultured GL-3 or transgenic plants were rooted and transferred to pots (30 cm × 18 cm) filled with equal parts of local loess sand and wormcast medium. Pots were placed in a greenhouse under natural illumination, with a temperature of 20–35°C and humidity of 35–55% for 1 month. Plants of each line were divided into a well-watered group ($n = 18$) and long-term water-deficit stress group ($n = 18$). Plants of the well-watered group were irrigated daily to maintain field capacity of 75–85%; plants of the long-term water-deficit stress group were daily irrigated to maintain a field capacity of 45–55%. Both treatments lasted for 2 months. At the end of treatment, leaves were harvested for morphology and physiological measurements.

Vector construction and transformation

For subcellular localization, the coding regions of *MdGH3.6* were cloned into the pEarleyGate 101 vector and fused to yellow fluorescent protein (YFP), resulting in YFP-MdGH3.6, which was then transformed into *Agrobacterium* C58C1. C58C1 carrying YFP-MdGH3.6 was transiently expressed in tobacco leaves according to the method described by Xie et al. (2018). YFP signals in transformed tobacco leaves were then detected with an FV1200 confocal microscope (Olympus, Tokyo, Japan).

In order to generate transgenic apple plants, CDS of *MdGH3.6* and *MdMYB94* were amplified by PCR and cloned into the pK2GW7 and pKGMYC vectors by gateway technology [Invitrogen (Carlsbad, CA, USA), Thermo Scientific (Waltham, MA, USA)]. For the RNAi construct, a 186-bp fragment of the *MdGH3* coding sequence was amplified by PCR and cloned into the pHELLSGATE2 vector by gateway technology (Invitrogen, Thermo Scientific, USA). *Agrobacterium*-mediated stable transformation of apple was carried out as described previously by Xie et al. (2018) using GL-3 as the genetic background and strain EHA105. For transient transformation, a 205-bp fragment of the *MdGH3.6* coding sequence was amplified by PCR and cloned into the pK7GWIWG2D vector by gateway technology (Invitrogen, Thermo Scientific, USA). *Agrobacterium*-mediated transient

transformation of apple was carried out as described previously (Zhang et al., 2018) using GL-3 as the genetic background and strain EHA105. The primers are listed in Table S1.

RNA isolation, RNA-seq analysis and RT-qPCR analysis

Plant leaves obtained from GL-3 and transgenic plants under control and water-deficit stress conditions were used for RNA extraction as described previously (Xie et al., 2018). RNAs were subjected to sequencing using the Illumina HiSeq platform by Novogene (Beijing, China) and aligned with the GDDH13 reference genome (Version 1.1) by Hisat2 (Kim et al., 2015). Differential gene expressions were analyzed by DESeq2 (Love et al., 2014) with threshold of *P*-value below 0.05 and the absolute value of log2 (fold-change) above 1. FPKM (fragments per kilobase of transcript per million fragments mapped) value was calculated by Cuffnorm in Cufflinks. Heatmaps were created by Tbtools (Chen et al., 2020a). RT-qPCR analysis was carried out as described (Xie et al., 2018). The primers are listed in Table S1.

IAA and metabolite measurements

The content of endogenous hormone IAA in the leaves was extracted from transgenic and GL-3 plants grown under the control and water-deficit stress conditions. The IAA was extracted and quantified according to the methods described by Lemos et al. (2016) with modifications. Briefly, 1 g of frozen leaf sample (three replicates) was ground into fine powder, and was extracted for 16 h with 10 ml 80% methanol including 0.5 g L^{-1} citric acid- H_2O and 0.2 g L^{-1} butylated hydroxytoluene (antioxidant) at 4°C under dark. Quantitative analysis of IAA content was performed using HPLC-MS/MS (QTRAP5500). The analytical conditions were as follows: HPLC column, Waters ACQUITY UPLC HSS T3 C18 (1.8 μm , 2.1 mm × 100 mm); solvent system, water (0.04% acetic acid):acetonitrile (0.04% acetic acid); gradient program, 75:25 v/v at 0–1 min, 5:95 v/v at 1.0–5.0 min, 5:95 v/v at 5.0–6.5 min, 75:25 v/v at 6.5–6.6 min, and 75:25 v/v at 6.6–10.0 min; flow rate, 0.70 ml min^{-1} ; temperature, 40°C; and injection volume, 5 μl . The effluent was alternatively connected to an ESI-triple quadrupole-linear ion trap (QTRAP)-MS. LIT and triple quadrupole (QQQ) scans were acquired by an API 6500 QTRAP LC/MS/MS triple quadrupole-linear ion trap mass spectrometer (QTRAP) system. The ESI source operation parameters were the same as those in the study by Lemos et al. (2016). The standard IAA was purchased from Sigma-Aldrich (St. Louis, MO, USA).

Metabolite measurements were performed according to previous methods (Chen et al., 2013) with modifications. Approximately 0.3 g of frozen leaf sample (three replicates) was ultrasonically extracted twice for 20 min with 3 ml methanol: deionized water:formic acid (25:24:1, v/v/v) at room temperature. Following centrifugation at 10 000 g min^{-1} for 15 min, the supernatant was filtered by a filter membrane with a pore size of 0.22 μm . The extracts were then analyzed using an HPLC-MS/MS system (QTRAP5500). The analytical conditions were as follows: HPLC column, Waters ACQUITY UPLC HSS T3 C18 (1.8 μm , 2.1 mm × 100 mm); solvent system, water (0.1% formic acid):100% methanol; gradient program, 95:5 v/v at 0.65–0.61 min, 5:95 v/v at 1.06–0.611 min, 5:95 v/v at 11.06–0.612 min, 95:5 v/v at 12.06–0.612 min, and 95:5 v/v at 12.16–0.615 min; flow rate, 0.40 ml min^{-1} ; temperature, 40°C; and injection volume, 2 μl . The effluent was alternatively connected to an ESI-triple quadrupole-linear ion trap (QTRAP)-MS. LIT and triple quadrupole (QQQ) scans were acquired by an API 6500 QTRAP LC/MS/MS triple quadrupole-linear ion trap mass spectrometer (QTRAP) system. The ESI source operation parameters were the same as those

in the study by Chen et al. (2013). The standard metabolites were purchased from Sigma-Aldrich (USA).

Quantification of leaf cuticular wax content

Cuticular wax was extracted and detected as described previously with modifications (Mao et al., 2012). Briefly, leaves of GL-3 and transgenic plants under control and long-term water-deficit stress conditions were immersed in 15 ml chloroform (HPLC grade) for 60 sec at room temperature, with 100 µg n-tetracosane as an internal standard. Chloroform extraction was filtered using 0.45 µm organic membrane (New Asia, Shanghai) and then evaporated using a pressure blowing concentrator (ZhiXin, Shanghai, MD200-2). The residue was derivatized with 40 µl of bis-N,N-(trimethylsilyl) trifluoroacetamide (Sigma-Aldrich) and 40 µl of pyridine for 60 min at 70°C. The derivative was then evaporated under gaseous N₂ and the dry substance was re-dissolved with chloroform, which was filtered using 0.22-µm organic membrane (New Asia, Shanghai) and analyzed with gas chromatography–tandem mass spectrometry (GC-MS) coupled with the Trace GC ULTRA/ISQ MS detector (Thermo Scientific, USA).

TRACE GC Ultra GC coupled with an ISQ mass spectrometer (Thermo Fisher Scientific, Waltham, MA, USA) was used to analyze 1 µl sample. The GC was fitted with a HP-INNOWAX column (60 m × 0.25 mm × 0.25 µm film thickness; Agilent, Palo Alto, USA). Helium was used as the carrier gas, with a split ratio of 10:1, at a flow rate of 1 ml min⁻¹. The temperature of the injection port, ion source and MS transfer line were 280°C, 240°C and 240°C, respectively. The temperature increasing process of the column chamber was as follows: initial temperature was 50°C for 2 min, followed by a ramp of 20°C min⁻¹ to 240°C for 2 min, then was increased to 320°C at a rate of 1.5°C min⁻¹, and finally kept at 320°C for 15 min. The MS was operated in positive electron ionization mode at 70 eV, obtaining spectra with a range of 406.5–6.6460 m/z. Cuticular wax compounds were quantified by the internal standard method. The total amount of cuticular wax was expressed per unit area of the leaf surface. Leaf area was measured using an EPSON Portable Area Meter (EPSON Perfection V19).

H₂O₂, enzyme activity and MDA measurements

H₂O₂ was detected as described previously (Xie et al., 2018). The activities of POD and CAT were measured as described previously (Wang et al., 2012). MDA content was determined according to the methods described previously (Sun et al., 2018).

Chlorophyll extraction

Chlorophyll content was determined as described previously (Mao et al., 2012) with modification. The leaves from GL-3 and transgenic plants under control and long-term water-deficit stress conditions were immersed in 15 ml 80% ethanol at room temperature. Aliquots of 0.2 ml were removed for chlorophyll quantification and returned to the same tube at the indicated time point. The chlorophyll concentration was quantified using a multi-function enzyme marker meter (Victor Nivo, PerkinElmer, Waltham, MA, USA) at wavelengths of 664 and 647 nm. Chlorophyll efflux at each time point was expressed as a percentage of total chlorophyll extracted after 24 h of immersion.

Ion leakage analysis

Ion leakage experiments were carried out using leaves of GL-3 and transgenic plants before or after water-deficit stress. Fully expanded leaves were punched into leaf discs with diameter of

50 mm. Six leaf discs were placed in a test tube containing 10 ml of deionized H₂O. After 4 h, the ionic exudation rate was measured by a conductivity meter and followed by another measurement after the leaf discs were autoclaved. Electrolyte leakage was calculated as the percentage of the conductivity before autoclaving relative to that after autoclaving (Verslues et al., 2006). Six leaves were used as one biological replicate, and six replicates were used for each treatment.

Dual-luciferase reporter assay system

The CDS of *MdMYB94* (MDP0000144744) and *MdMYB96* (MDP0000149535) were cloned into the pGreenII 62-SK vector, which was used as an effector. The empty vector of pGreenII 62-SK was used as a negative control. Promoter sequences (601 bp) of *MdGH3.6* were inserted into the pGreenII 0800-LUC vector, which was used as a reporter. The primers are listed in Table S1.

The binary constructs were transformed into *Agrobacterium* strain GV3101 by heat shock transformation. The vector-transformed *Agrobacterium* was cultured at 28°C with shaking at 200 rpm until OD₆₀₀ = 1.0. The *Agrobacterium* was then suspended in liquid medium [10 mM MES-KOH (pH 5.2), 10 mM MgCl₂ and 100 µM acetosyringone] and incubated for 2 h before coinfiltration into *Nicotiana benthamiana*. Firefly luciferase (LUC) and Renilla luciferase (REN) activity levels were measured using the Dual-Luciferase Reporter Assay System (Promega, NO.E1910, Madison, WI, USA). The LUC activity was normalized to REN activity. Twelve biological replicates were performed for this analysis.

DAP-qPCR assay and ChIP-qPCR assay

The CDS of *MdMYB94* and *MdMYB96* were cloned into the pET-42a vector. Using the TNT SP6 High-Yield Wheat Germ Protein Expression System (Promega, NO.L3260, USA), these cDNAs were transcribed and translated *in vitro*. DAP-qPCR and ChIP-qPCR were performed as described previously (Chen et al., 2020b; Xie et al., 2018). The primers are listed in Table S1.

EMSA

The EMSA was performed as described by Xie et al. (2018). The CDS of *MdMYB94* was cloned into the pMAL-C5X vector. The resulting plasmid was then expressed in the *Escherichia coli* BL21 (DE3) to produce recombinant MBP-MYB94 protein. The Light-Shift™ Chemiluminescent EMSA Kit (Thermo Scientific, NO. 20148, USA) was used for the binding reactions. The probes labeled with biotin at the 5'-end of single strand were synthesized by Shanghai Sangon Biotech (Shanghai, China). Primers and oligonucleotide probe sequences are listed in Table S1.

Statistical analysis

Data are reported as the mean ± SD. Statistical significance was determined by one-way ANOVA (Tukey's test) analysis using SPSS (version 21.0, IBM, Armonk, NY, USA). Variations were considered significant if *P* < 0.05, 0.01 or 0.001.

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AUTHOR CONTRIBUTIONS

QG conceived and designed the project; LJ, CL, WS and QY performed experiments; JH and FM analyzed data; DZ, LJ and QG wrote the manuscript; NC, FY, XL, XS, NH and PC helped with the discussion of the work. All the authors contributed to the preparation of the final manuscript.

CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

DATA AVAILABILITY STATEMENT

RNA-seq data in this study have been submitted to NCBI under the project of PRJNA645516.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article.

Figure S1. Subcellular localization of MdGH3.6.

Figure S2. Expression of *MdGH3.6* in *MdGH3.6* overexpression transgenic plants.

Figure S3. Phenotype of *MdGH3.6* OE plants.

Figure S4. Photosynthetic parameters of GL-3 and *MdGH3.6* OE plants in response to water-deficit stress.

Figure S5. IAA content in GL-3 and *MdGH3.6* OE plants under control and water-deficit stress conditions.

Figure S6. Protein and CDS alignment of apple MdGH3s analyzed in this study. CDS, coding region sequence.

Figure S7. Expression pattern of apple *GH3.6* like and *MdGH3.1* genes in response to water-deficit stress.

Figure S8. Expression of *MdGH3* genes in *MdGH3.6* RNAi transient transgenic plants.

Figure S9. Expression of *MdGH3* genes in *MdGH3* RNAi transgenic plants.

Figure S10. Phenotype of *MdGH3* RNAi plants.

Figure S11. Photosynthetic parameters of GL-3 and *MdGH3* RNAi under water-deficit stress.

Figure S12. IAA content in GL-3 and *MdGH3* RNAi plants under control and water-deficit stress conditions.

Figure S13. Heat map showing the log₂(fold-change) in transcript level of downregulated and upregulated genes shown in *MdGH3* RNAi plants under water-deficit stress conditions.

Figure S14. The enriched GO terms for upregulated genes shown in *MdGH3* RNAi plants under water-deficit stress conditions.

Figure S15. The enriched GO terms for downregulated genes shown in *MdGH3* RNAi plants under water-deficit stress conditions.

Figure S16. KEGG enrichments for downregulated genes shown in *MdGH3* RNAi plants under water-deficit stress conditions.

Figure S17. Expression of other wax biosynthesis-related genes in GL-3, *MdGH3* RNAi or *MdGH3.6* OE plants in response to water-deficit stress.

Figure S18. Chlorophyll extraction from GL-3 and *MdGH3* RNAi plants under control and long-term water-deficit stress conditions.

Figure S19. Cuticular wax content of GL-3 and *MdGH3.6* OE plants under water-deficit stress.

Figure S20. MdMYB96 can not bind to *MdGH3.6* promoter by a dual-luciferase assay.

Figure S21. Regulation of MdMYB94 expression by water-deficit stress.

Figure S22. *MdMYB94* OE calli are more tolerant to simulated drought stress.

Figure S23. Expression of *MdMYB94* in *MdMYB94* overexpression transgenic plants.

Figure S24. Protein alignment of apple MdGH3.6 and Arabidopsis GH3.6.

Table S1 Primers used in this study

Dataset S1. Genes induced by water-deficit stress in GL-3.

Dataset S2. Genes repressed by water-deficit stress in GL-3.

Dataset S3. Genes upregulated in RNAi plants under control conditions.

Dataset S4. Genes downregulated in RNAi plants under control conditions.

Dataset S5. Genes upregulated in RNAi plants under water-deficit stress.

Dataset S6. Genes downregulated in RNAi plants under water-deficit stress.

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